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The below program illustrates nested switch statement example. The outer switch checks for zero or non-zero and the inner switch checks for odd or even.

Illustration 10.10 C++ program to check for zero or non-zero and odd or even using nested switch statement

```
#include <iostream>
using namespace std;
int main()
{
int a = 8;
cout<<"The Number is : " <<a <<endl;
switch (a)
        {
        case 0 :
        cout<<"The number is zero" <<endl;
        break;
        default:
        cout<<"The number is a non-zero integer" <<endl;
        int b = a % 2;
        switch (b)
{
        case 0:
        cout<<"The number is even" <<endl;
        break;
        case 1:
        cout<<"The number is odd" <<endl;
        break;
        }
}
        return 0;
}
```

Output

```
The Number is : 8
The number is a non-zero integer
The number is even
```

10.5 Iteration statements

An iteration (or looping) is a sequence of one or more statements that are repeatedly executed until a condition is satisfied. These statements are also called as control flow statements. It is used to reduce the length of code, to reduce time, to execute program and takes less memory space. C++ supports three types of iteration statements;